Name: Data Science Assignment

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(1) (a) What is Data Mining? (b) Explain how the evolution of database technology led to data mining.

Ans (a) Data Mining is a process of searching, extracting & analysing useful information in large data repositories. Data Mining techniques are deployed to scour large databases in order to find novel & useful patterns that might otherwise remain unknown.

They also provide capabalities to predict the outcome of a future observations such as predicting whether a newly arrived customer will spend more than $100 at a department store.

b) Data Mining can be viewed as a result of the natural evolution of information technology. As the database & data management industry evolved in the development of several critical functionalities such as from Data Collection and Database Creation (1960s & earlier to DBMS (1970s to early 1980s) to Advanced Database System (mid 1980s to present) & and Advanced Data Analysis (late-1980s to present)

(Data Mining):

The abundance of data, coupled with the need for powerful data analysis tools, has been described as a data rich but information poor The growing, tremendous amount of data, collected & stored in large & numerous data repositories, has led to a situation called data tombs (data archives that are seldom visited).

Efforts have been made to develop expert system & knowledge based technologies, which typically rely on users or domain experts to manually input the knowledge bases.

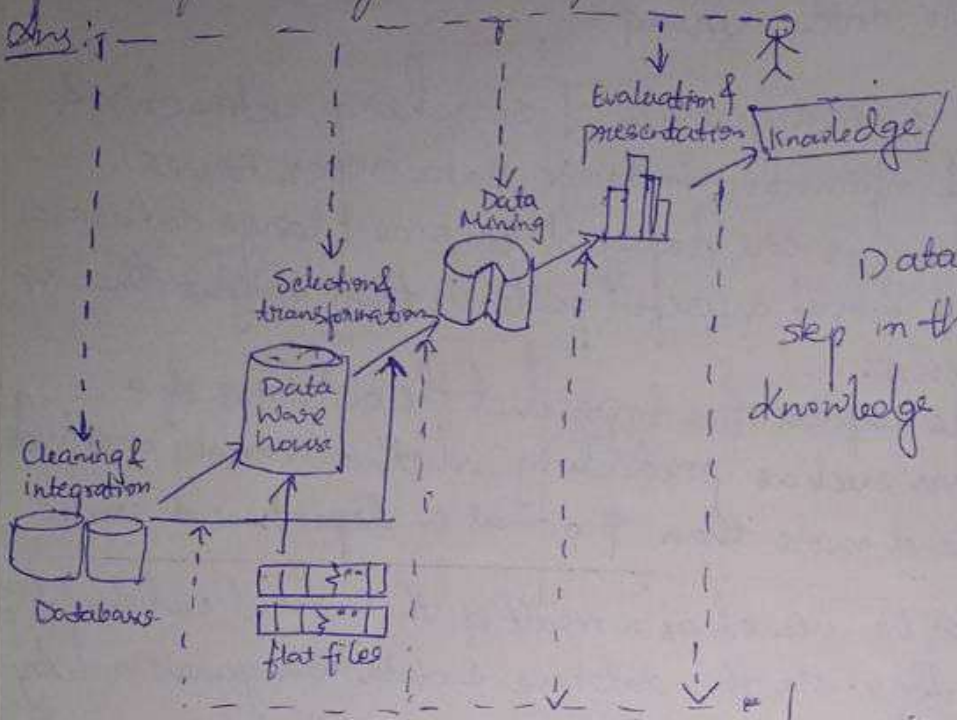
Unfortunately, however, the manual knowledge input procedure is prone to biases & errors & is extremely costly & time consuming.

The widening gap between data & information calls for the systematic

development of data mining tools that can turn data into "golden nuggets" of knowledge

2. Describe the steps involved in data mining when viewed as a process of knowledge discovery?

Ans:



Data Mining as a step in the process of knowledge discovery

The steps involved in data mining when viewed as a process of Knowledge discovery.

1) Data cleaning :- In this step the noise & inconsistent data is removed

2) Data Integration :- Here, multiple data sources are combined

3) Data Selection :- Here, data relevant to the analysis task are retrieved from the database.

4) Data transformation :- In this step, data is transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations

5) Data Mining :- In this step, data patterns are extracted using intelligent methods

6) Pattern evaluation :- It is used to identify the truly interesting patterns representing knowledge based on interesting measures.

7)

4) Knowledge presentation : ~~Knowledge is presented using~~ Here visualization & knowledge representation techniques are used to present mined knowledge to the users.

Here, Steps 1 to 4 are different forms of data preprocessing, where data are prepared for mining. The data mining step interacts with the user or a knowledge base.

(3) How is a data warehouse different from a database? How are they similar?

Ans: Difference

Difference	Database	Data Warehouse
Definition	Database is an organized collection of data stored, manipulated & retrieved as per requirement	Datawarehouse is ~~for analysis~~ analysis & generating reports due to vast range & different types of data.
Design	Design of operational database is different from data warehouse design. It mainly observes data accuracy when updating real time data	The design of data warehouse ensures vast range of data which is used over time for analysis purpose.
Focus	The focus of database is mainly on transactions with the help of queries	It has ability to of data analytics which is collected from different sources & generate reports
Type of information	The databases provide information such as online availability of seats	The information (analysis report) obtained from data warehouse are used for betterment & to check the performance of business
Types	There are many types of databases. The examples are OLTP, CSV, text files, excel, spreadsheets & XML files etc.	It is an OLAP type of database which exist on the top layer of other database and perform analysis
Optimization	It is optimized for read-write operations through single point transaction. Mostly the OLTP database queries respond in less than a second	The data warehouses are optimized for retrieval of large datasets to aggregate the data as it is designed for handling broad analytical queries

Difference	Database	Data Warehouse
Reporting	It has more static type of reporting. These are usually one time lists. For example, you may need data of a specific patient to check the history of disease or a student checks its admission status. These results can be in PDF formats. The data may be combined from several tables & actual queries may be complex which require expertise in this field	When you measure the performance of a business, the static reports is not sufficient for that purpose. To perform analysis, the data is aggregated & summarized into different types of reports.
Data Duplication	In an OLTP database, the data is normalized & there is no duplication of data in order to increase the optimized processing & better efficiency	In OLAP database, the data is organized in such a way to facilitate the analysis & reporting. Usually the data is denormalized & stored in fewer tables in simple structure

Similarities

~~Similarities~~ Database and Data Warehouse

1) Database & Data Warehouse both the systems maintain data in form of table, indexes, columns, views & keys.
2) Data is retrieved in both by using SQL queries
3) Both are used for storing data
4) Data Warehouse bottom tier is a Relational database System even databases are also relational database system
5) Both supports multi user access. A single instance of
6) database & data warehouse can be accessed by many users at a time.
6) Both DB & DW can be present on cloud or local
7) A data warehouse is also a database

(Q) List and describe the five primitives for specifying a data mining task.

Ans: The data mining primitives are:-

1) Task-relevant data: This is the database position to be investigated For example, suppose that you are a manager of All Electronics in charge of sales in the US & Canada. In particular, you would like to study the buying trends of customers in Canada. Rather than mining on the entire database. These are referred to as relevant attributes.

2) The kinds of knowledge to be mined: This specifies the data mining functions to be performed, such as characterization, discrimination, association, classification, clustering or evolution analysis. For instance, if studying the buying habits of customers in Canada, you may choose to mine associations between customer profiles & the items that these customers like to buy.

3) Background Knowledge: Users can specify background knowledge, or knowledge about the domain to be mined. This knowledge is useful for guiding the knowledge discovery process, & for evaluating the patterns found. There are several kinds of background knowledge.

4) Interestingness measures: These functions are used to separate uninteresting patterns from knowledge. They may be used to guide the mining process, or after discovery, to evaluate the discovered patterns. Different kinds of knowledge may have different interestingness measures

5) Presentation & visualization of discovered patterns This refers to the form in which discovered patterns are to be displayed. Users can choose from different forms for knowledge presentation, such as rules, tables

charts, graphs, decision trees & cubes.

Task relevant data
- Database or datawarehouse name
- database tables or datawarehouse cubes
- conditions for data selection
- relevant attributes or dimensions
- data grouping criteria

Knowledge type to be defined
- characterization
- discrimination
- association
- classification / prediction
- clustering

Background knowledge
- concept hierarchies
- user beliefs about relationships in the data.

Pattern - interestingness measurements
- simplicity
- certainity (Ex:- confidence)
- utility (Ex:- support)
- novelty

Visualization of discovered patterns
- rules, tables, reports, charts, graphs, decision trees & cubes
- drill down & roll up.

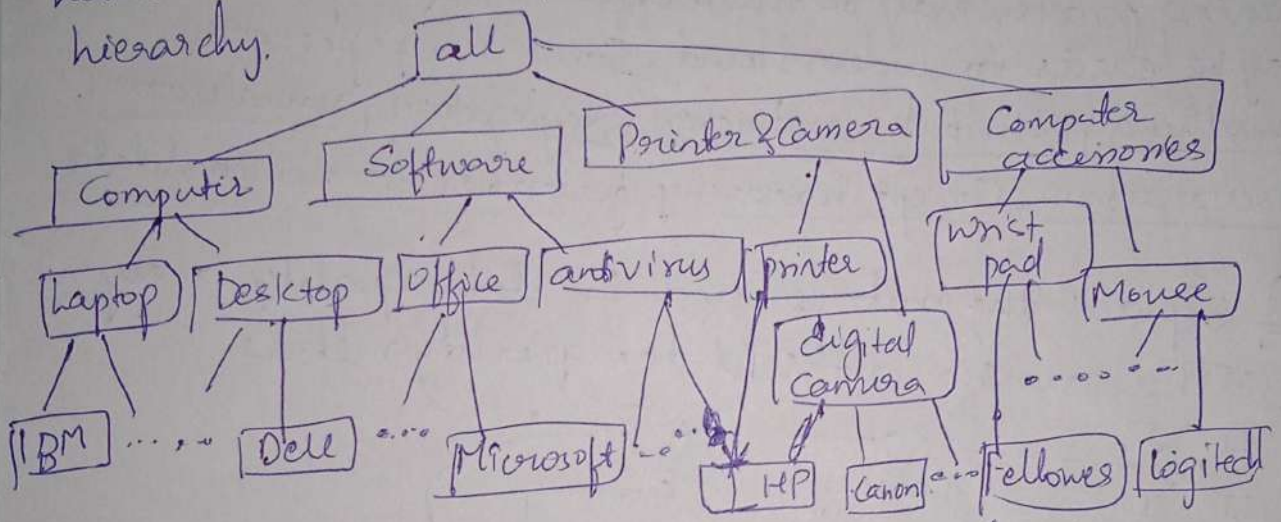
(5) Describe ~~the~~ why concept hierarchies are useful in data mining?

Ans:- Concept hierarchies are used to organize attributes or attribute values into diff levels of abstraction.
-> Concept hierarchies are used to generalize the data where low level ~~concepts through the use~~ or primitive (raw) data are replaced by higher level concepts ~~through~~ t
-> Concept hierarchies allow the mining of data at multiple levels of abstraction, & are a powerful tool for data mining

→ Multilevel association rules can be mined efficiently using concept hierarchies under a support-confidence framework.

→ The concepts hierarchy has five levels, respectively referred to as levels 0-4 starting with level 0 at the root node for all

- Level 1 includes computer, s/w, printer, camera & computer accessories.
- Level 2 includes laptop computer, desktop computer, office s/w, antivirus s/w
- Level 3 includes IBM desktop computer, Microsoft office s/w & so on
- Level 4 is the most specific abstraction level of this hierarchy.



(6) Discuss data pre-processing tasks in detail

Ans: 1) Data Cleaning: ~~It is a process~~ of It is used to cleanse the data by filling in missing values, smoothing noisy data, identifying or removing outliers, & resolving inconsistencies.

2) Data Integration: It is used to combine data from multiple sources into a coherent data store, as in data warehousing. These sources may include multiple databases, data cubes or flat files. It includes Entity identification problem, Redundancy & Correlation analysis, Tuple duplication and Data Value Conflict detection & resolution

3) ~~Data Reduction~~: -

(3) Data Reduction: - These techniques can be applied to obtain a reduced representation

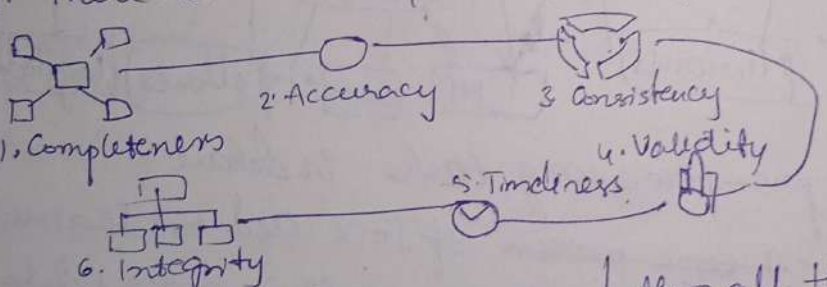
of the dataset that is much smaller in volume, yet closely maintains the integrity of the original data. That is, mining on the reduced data set should be more efficient yet produce the same analytical results. Some techniques are Dimensionality reduction, Numerosity reduction, data compression, wavelet transforms, Principal component Analysis (PCA), Attribute subset selection, Regression & log linear models, histograms, Clustering, Sampling, Data Cube Aggregation,

(4) Data Transformation and discretization

The data are transformed or consolidated so that the resulting mining process may be more efficient, & the patterns found may be easier to understand. Some strategies are Smoothing, Attribute construction, Aggregation, Normalization, Discretization, Concept hierarchy generation for nominal data

(7) Explain the metrics used in data quality analysis?

Ans:- There are six different dataquality metrics



1) Completeness :- This measures whether all the necessary data is present in a specific dataset. You can think about completeness in one of the 2 ways : at the record level or at the attribute level. Measuring completeness at the attribute level is a little more complex however, as not all fields will be mandatory. An example metric for completeness is the percent of data fields that have values entered into them.

2) Accuracy :- How accurately does your data reflect the real-world object ? In the financial sector, data accuracy is usually black or white - it either is or isn't accurate. That's because the number of pounds & pennies in an acc is a precise number

Data accuracy is critical in large organizations, where the penalities for failure are high.

5) Consistency: Maintaining synchronicity between different database - is essential. To ensure data remains consistent on a daily basis, software systems are often the answer. For ex:- transaction consistency helps you detect incomplete financial transactions. Once found, you can roll back those errors meaning both balances remain in check.

4) Validity: - It is a measure of how well data conforms to required value attributes. For ex:-, ensuring dates conform to the same format i.e, date/months/year or month/date/year

5) Timeliness: It reflects the accuracy of data at a specific point in time. An example of this is when a customer moves to a new house, how timely are they in informing their bank of their new address? Few people do this immediately, so there will be a negative infact on the timeliness of their data. Poor timeliness can also lead to bad decision making.

6) Integrity: To ensure data integrity, it is important to maintain all the data qualitymetrics ~~we've met~~. Typically data stored in multiple systems breaks data integrity. An example metric for integrity is the percent of data that is the same pacross multiple systems.

8) Describe various methods for handling tuples with missing values for some attributes

Ans The various methods for handling tuples with missing values are.

1) Ignore the tuple: This is usually done when the class label is missing. This method is not very effective, unless the tuple contains several attributes with missing values. It is especially poor when the ~~percentage~~ of missing values per attribute varies considerably. By ignoring the tuple, we do not make use of the remaining attributes values in the tuple. Such data could have been useful to the task at hand.

2) Fill in the missing value manually : In general, this approach is time consuming & may not be feasible given a large dataset with missing values

3) Use a global constant to fill the missing value :- Replace all the missing attribute values by the same constant such as a label like "Unknown" or -∞. If missing values are replaced by say, "Unknown", then the mining program may mistakenly think that they form an interesting concept, since they all have a value in common - that of "Unknown". Hence, although the method is simple, it is not foolproof.

4) Use a measure of central tendancy for the attribute (Ex:- the mean or median) to fill in the missing value :- For normal distribution, the mean can be used, while skewed data distribution should employ median. For example, suppose the data distribution regarding the income of customers is symmetric & that the mean income is 56000. Use this value to replace.

5) Use the attribute mean or median for all samples belonging to the same class as the given tuple :-

For example :- Suppose if classifying customers according to credit risk, we may replace the missing value with the mean income value for customers in the same credit risk category as that of the given tuple. If the data distribution for a given class is skewed, the median value is a better choice.

6) Use the most probable value to fill in the missing value

This may be determined with regression, inference-based tools using Bayesian formalism, or decision tree induction. For example, using the other customer attributes in your dataset you may construct a decision tree to predict the missing values for income.

(9) Why data cleaning is essential ?

Ans. Data cleaning :- Real world data tend to be incomplete noisy & inconsistent. Data cleaning / cleansing routines attempt

roach to fill in missing values, smooth out noise while identifying outliers set & correct inconsistencies in the data.

Data cleansing is important becoz it improves your data quality & in doing so, increases overall productivity. When you clean your data, all outdated or incorrect information is gone, leaving you with highest quality information

The data cleansing process involves reviewing all the data present within a database to either remove or update information that is incomplete, incorrect or duplicated & irrelevant.

Data cleansing is just not simply about erasing the old information to make space for new data, but the process is about rather finding a way to maximize the dataset's accuracy w/o necessarily tampering with the data available. Data cleaning is the process of determining & correcting the wrong data

Some data cleaning tasks are.

a) Missing values :- i) Ignore the tuple.
ii) Fill in the missing value manually
iii) Use a global constant to fill in the missing value
iv) Use a measure of central tendency for the attribute
v) Use the attribute mean or median for all samples belonging to the same class as the given tuple.
vi) Use the most probable value to fill in the missing value.

b) Noisy Data :- i) Binning, (ii) Regression (iii) Outlier analysis

c) Data Cleaning as process

(10) Explain Data Integration?

Ans:- The merging of data from multiple data stores. Careful integration can help reduce and avoid redundancies & inconsistencies in the resulting dataset. This can help improve the accuracy and speed of the subsequent data mining process

Some data integration tasks are:-

i) Entity identification problem: When matching attributes from one database to another during integration, special attention must be paid to the structure of the data. There may be number of issues to consider during data integration.

Schema integration & object matching will be tricky. How can equivalent real world entities from multiple sources can be matched? This should be ensured that any attribute, functional dependencies & referential constraints in the source system match those in the target system.

2) Redundancy and Correlation Analysis: Redundancy is another important issues in data integration. Inconsistencies in attribute or dimension naming can also cause redundancies in the dataset. Some redundancies can be detected by correlation analysis. Given 2 attributes such analysis can measure how strongly one attribute implies the other, based on the available data. For nominal data we use chi-square test. For numeric data we use the correlation coefficient & covariance both of which access how one attribute's values vary from those of another.

3) Tuple duplication: In addition to detecting redundancies between attributes, duplication should also be detected at the tuple level. The use of denormalized tables is another source of data redundancy. Inconsistencies often arise between various duplicates, due to inaccurate data entry or updating some but not all data occurrences.

4) Data Value Conflict Detection & Resolution: Data integration also involves the detection & resolution of data value conflicts. For example, for the same real world entity, attribute values from different sources may differ. This may be due to differences in representation, scaling or encoding. For instance, a weight attribute may be stored in metric unit in one system & British imperial units in another.

(11) Discuss Data Reduction Techniques?

Ans:- Data reduction techniques can be applied to obtain a reduced representation of the dataset that is much smaller in volume, yet closely maintains the integrity of the original data. That is, mining on the reduced dataset should be more efficient yet produce the same analytical results.

Dimensionality reduction, Numerosity reduction & data compression are the techniques used

1) Wavelet transforms: The discrete wavelet transform is a linear signal processing technique that, when applied to a data vector X, transforms it to a numerically different vector, X', of wavelet coefficients. The two vectors are of the same length.

2) Principal Components Analysis (PCA): Suppose data to be reduced consist of tuples or data vectors described by n attributes or dimensions PCA or K-L (Karhunen method) searches for k n-dimensional orthogonal vectors that can best be used to represent the data, where $k \leq n$. The original data are thus projected onto a smaller space, resulting in dimensionality reduction.

3) Attribute subset selection: Datasets for analysis may contain hundreds of attributes many of which may be irrelevant to the mining task or redundant. For example, if the task is to classify customers based on whether or not they are likely to purchase a CD when notified for sale, attributes like customer's telephone number are likely to be irrelavant.

4) Regression and log linear models: Regression & log linear models can be used to approximate the given data. In linear regression the data are modelled to fit a straight line. For ex: $y = mx + b$. log linear models approximate discrete multidimensional probability distributions. Given a set of tuples in n dimensions we consider each tuple as a point in an n-dimensional space. log linear models can be used to estimate the probability of each point in a multidimensional space for a set of discretized attributes based on smaller subset of dimensional combinations

5) Histograms: They use binning to approximate data distribution & are a popular form of data reduction. A histogram for an attribute A, partitions the data distribution of A into disjoint subsets, referred to as buckets or bins. If each bucket or bins represents only a single attribute value/frequency pair, the buckets are called singleton buckets.

6) Clustering: This technique considers data tuples as objects. They partition the objects into groups, or clusters, so that

object ~~in other~~ within a cluster are 'similar' to one another ~~and~~ and 'dissimilar' to objects in other clusters. Similarity is commonly defined in terms of how 'close' the objects are in space, based on distance formula. Centroid distance is an alternative ~~dis~~measure for cluster quality.

7) Sampling :- It can be used as a data reduction technique becoz it allows a large data set to be represented by a much smaller random data sample (or subset). Suppose that a large dataset D, contains N tuples ~~tetwe~~ could sample D for data reduction.

8) Data Cube Aggregation :- Suppose you have collected the data for your analysis. These data consists of the sales per quarter, for the years 2008 to 2010, However we are interested in the annual sales, rather than total per quarter. Thus the data can be aggregated so that the resulting data summarize the total sales per year instead of quarter. The resulting dataset is smaller in volume, w/o loss of information.

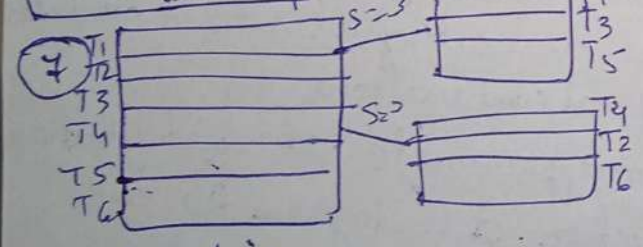
Year	2010
Quarter	Sales

Year	2009
Quarter	Sales

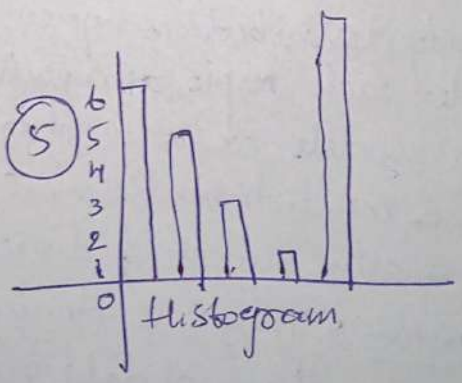
Year	2008
Quarter	Sales
Q1	224000
Q2	408000
Q3	350000
Q4	586000

=>

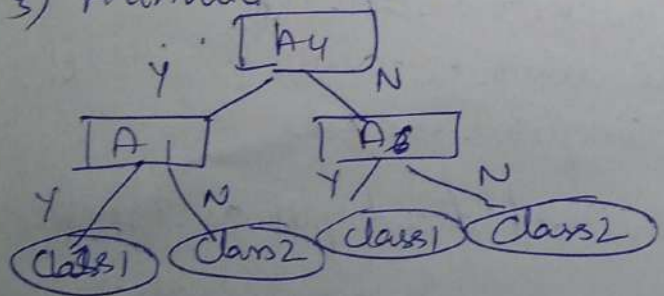
Year	Sales
2008	1568000
2009	2356000
2010	3594000



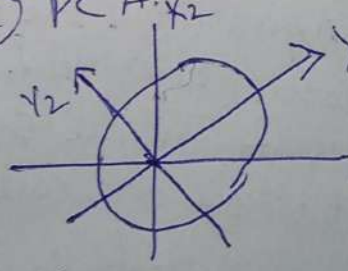
Sampling



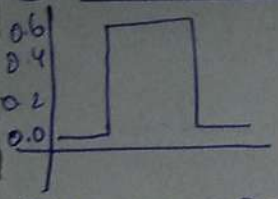
3) Attribute subset selection



(2) PCA.



(1) Wavelet transform



(12) Explain Data Transformation Strategies ?

Ans :- The data are transformed or consolidated. so that the resulting mining process may be more efficient, & the patterns found may be easier to understand. Data discretization, a form of data transformations is

1) Smoothing :- which works to remove noise from the data. Techniques include binning, regression & clustering

2) Attribute construction :- where new attributes are constructed & added from the given set of attributes to help the mining process

3) Aggregation :- where summary or aggregation operations are applied to the data. For example :- the daily sales data may be aggregated so as to compute monthly & annual total amounts. This step is typically used in constructing a data cube for data analysis at multiple abstraction levels

4) Normalization :- where the attribute data are scaled so as to fall within a smaller range, such as -1.0 to 1.0 or 0.0 to 1.0

5) Discretization :- where the raw values of a numeric attribute are replaced by interval labels (e.g. 0-10, 11-20 etc) or conceptual labels (e.g., youth, adult, senior). The labels, in turn, can be recursively organized into higher level concepts, resulting in a concept hierarchy for the numeric attribute.

6) Concept hierarchy generation for nominal data :- where attributes such as street can be generalized to higher level concepts, like city or country. Many hierarchies for nominal attributes are implicit within the database schema & can be automatically defined at the schema definition level.

(15) For the attribute age: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

(1) Use smoothing by bin means to smooth the above data, using a bin depth of 3 illustrate your steps.

bin depth = 3

bin size => no. of bins

Step 1: Sort the data (already sorted)

Step 2: Using bin depth 3 we will create bins

Bin 1 - 13 15 16
Bin 2 - 16 19 20
Bin 3 - 20 21 22
Bin 4 - 22 25 25
Bin 5 - 25 25 30
Bin 6 - 33 33 35
Bin 7 - 35 35 35
Bin 8 - 36 40 45
Bin 9 - 46 52 70

Smoothing by bin means: take the mean of each bin

Bin 1 = $\frac{13+15+16}{3}$ => 15, 15, 15

Bin 2 = $\frac{16+19+20}{3}$ => 18, 18, 18

Bin 3 = $\frac{20+21+22}{3}$ => 21, 21, 21

Bin 4 = $\frac{22+25+25}{3}$ => 24, 24, 24

Bin 5 = $\frac{25+25+30}{3}$ => 26.6 -> 27, 27, 27

Bin 6 = $\frac{33+33+35}{3}$ => 33.6 => 34, 34, 34

Bin 7 = $\frac{35+35+35}{3}$ => 35, 35, 35

Bin 8 = $\frac{36+40+45}{3}$ => 40.3 => 40, 40, 40

Bin 9 = $\frac{46+52+70}{3}$ = 56, 56, 56

(2) Use min-max normalization to transform the value 35 for age into the range [0.0, 1.0].

Formula = $v_i' = \frac{v_i - min_A}{max_A - min_A}(new_max_A - new_min_A) + new_min_A$

$v_i = 35$, $min_A = 13$, $max_A = 70$, $new_max_A = 1.0$, $new_min_A = 0.0$

$v_i' = \frac{35-13}{70-13}(1.0-0.0)+0.0$ =>

$= \frac{22}{57}(1.0)$

$v_i' = 0.38596$

3) Use z-score normalization to transform the value 35 for age, where the standard deviation of age is 12.94 years.

Formula $v_i' = \frac{v_i - \bar{A}}{\sigma_A}$ ~~$S_A = \frac{1}{n}(|v_1 - \bar{A}| + |v_2 - \bar{A}| \ldots)$~~.

$\sigma_A = 12.94$

$\bar{A}$ = mean of A $\Rightarrow \frac{809}{27} = 29.96$

$v_i' = \frac{35 - 29.96}{12.94} = 0.3894$

(4) Use normalization by decimal scaling to transform the value 35 for age.

The maximum value is 70 so we will choose j as 2.

Any - Formula $x' = \frac{x}{10^j}$

$\therefore x' = \frac{35}{10^2} = 0.35$

(5) Given data: 200, 300, 400, 600, 1000. Normalize the data using the methods.

1) Min max normalization by setting min = 0 & max = 1

Formula $v_i' = \frac{v_i - min_A}{max_A - min_A} \times (new_max_A - new_min_A) + new_min_A$

~~$v_i = min_A$~~ $= 200$ $max_A = 1000$ $new_max_A = 1$ $new_min_A = 0$

$v_1 = 200 \rightarrow v_1' = \frac{200 - 200}{1000 - 200}(1 - 0) + 0 = 0$

$v_2 = 300 \rightarrow v_2' = \frac{300 - 200}{1000 - 200}(1) = 0.125$

$v_3 = 400 \rightarrow v_3' = \frac{400 - 200}{1000 - 200} = 0.25$

$v_4 = 600 \rightarrow v_4' = \frac{600 - 200}{1000 - 200} = 0.5$

$v_5 = 1000 \rightarrow v_5' = \frac{1000 - 200}{1000 - 200}(1) = 1$

2) Use z score normalization.

$v_i' = \frac{v_i - \bar{A}}{\sigma A}$ $\sigma A = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2}$ $\mu = \bar{A} = 500$

$$\sigma_A = \sqrt{\frac{1}{5}\left[(200-500)^2+(300-500)^2+(400-500)^2+(600-500)^2+(1000-500)^2\right]}$$

$$= \sqrt{\frac{1}{5}(90{,}000+40{,}000+10000+10000+250000)}$$

$$\sigma_A = \sqrt{80{,}000} \Rightarrow 282.8427$$

$v_1 = 200 \quad v_1' = \frac{200-500}{282.8427} = -1.060$

$v_2 = 300 \quad v_2' = \frac{300-500}{282.8427} = -0.7071$

$v_3 = 400 \quad v_3' = \frac{400-500}{282.8427} = -0.35355$

$v_4 = 600 \quad v_4' = \frac{600-500}{282.8427} = 0.35355$

$v_5 = 1000 \quad v_5' = \frac{1000-500}{282.8427} = 1.7677.$

(3) z-score normalization using mean absolute deviation instead of standard deviation.

$$F = v_i' = \frac{v_i - \bar{A}}{S_A} \qquad S_A = \frac{1}{n}(|v_1-\bar{A}| + |v_2-\bar{A}| \cdots)$$

$\bar{A} = 500$

$$S_A = \frac{1}{5}(|200-500|+|300-500|+|400-500|+|600-500|+|1000-500|)$$

$$S_A = \frac{1}{5}(300+200+100+100+500) = 240$$

$v_1 = 200 \quad v_1' = \frac{200-500}{240} = -1.25$

$v_2 = 300 \quad v_2' = \frac{300-500}{240} = -0.833$

$v_3 = 400 \quad v_3' = \frac{400-500}{240} = -0.416$

$v_4 = 600 \quad v_4' = \frac{600-500}{240} = 0.416$

$v_5 = 1000 \quad v_5' = \frac{1000-500}{240} = 2.083$

(4) normalization by decimal scaling

$F = x' = \frac{x}{10^j}$ Max value of the given sequence is 1000

So $j = 4$

$x_1 = 200 \quad x_1' = \frac{200}{10^4} = 0.02$

$x_2 = 300 \Rightarrow x_2' = \frac{300}{10^4} = 0.03$

$x_3 = 400 \Rightarrow x_3' = \frac{400}{10^4} = 0.04$

$x_4 = 600 \rightarrow x_4' = \frac{600}{10^4} = 0.06$

$x_5 = 1000 \rightarrow x_5' = \frac{1000}{10^4} = 0.1$

if $j = 3$ then $x_1' = 0.2$ as they all lie inside 10^3

$x_2' = 0.3$

$x_3' = 0.4$

$x_4' = 0.6$

$x_5' = 1$

(15) Survey of 2000 coffee drinkers is given below. Find the Chi-square value

Response	% of coffee drinkers	Observed frequency
2 cups per week	20 %	306
1 Cup per week	15 %	293
2 Cups per day	27 %	562
1 cup per day	38 %	839

Chisquare X^2

Let H_0 : The population frequencies are equal to the expected frequencies

H_a : The population freq are not equal to expected freq

The degrees of freedom $= k - 1 = 4 - 1 = 3$

Response	% of Coffee drinkers	E	O	O − E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
2 cups per week	20 %	0.20 × 2000 = 400	306	−94	8836	22.09
1 cup per week.	15 %	300	293	−7	49	0.163
2 cups per day	27 %	540	562	22	484	0.8962
1 cup per day	38 %	760	839	79	6241	8.211

$\chi^2 = \Sigma \frac{(O-E)^2}{E} \Rightarrow 31.3602$ is the chi square value